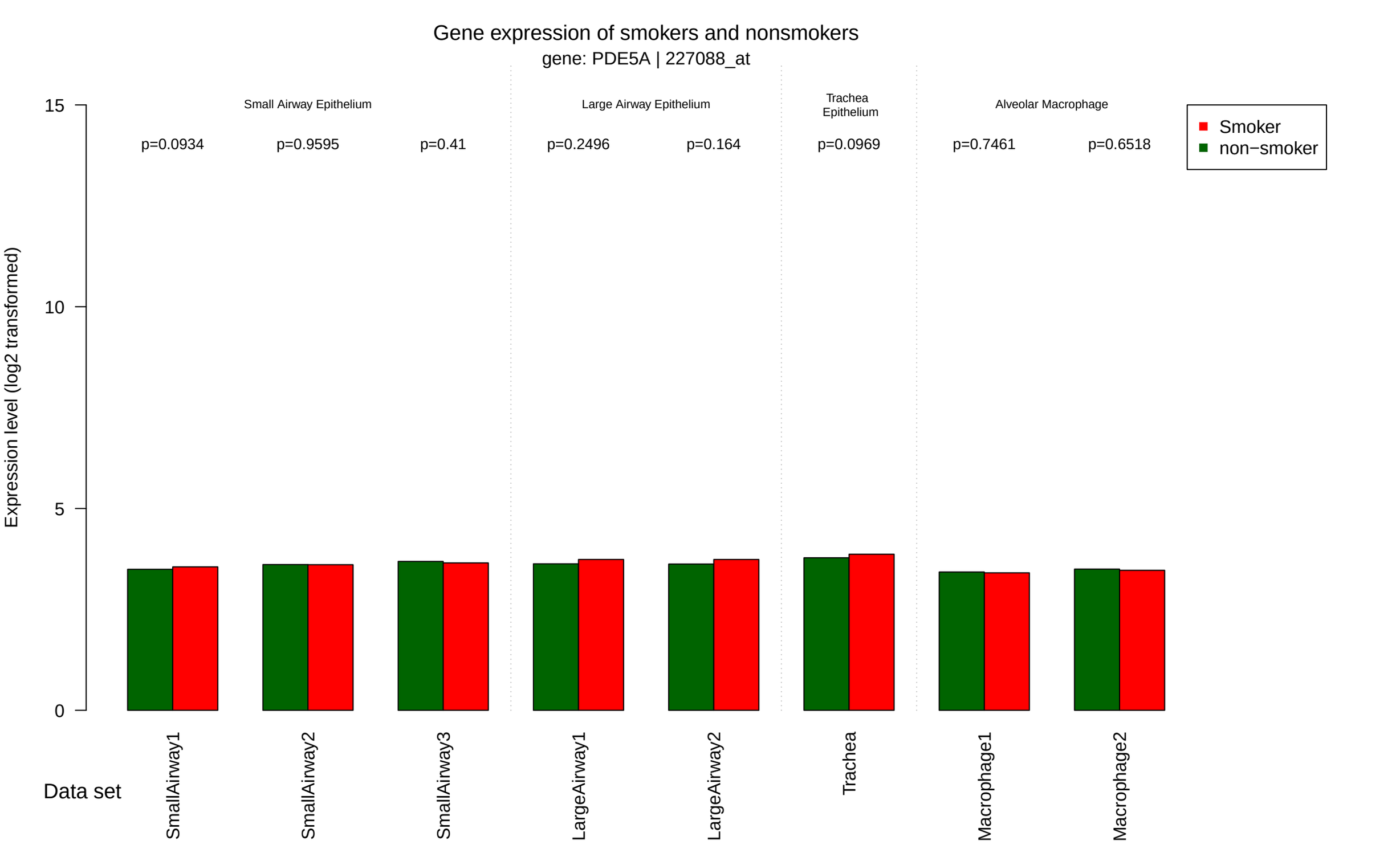
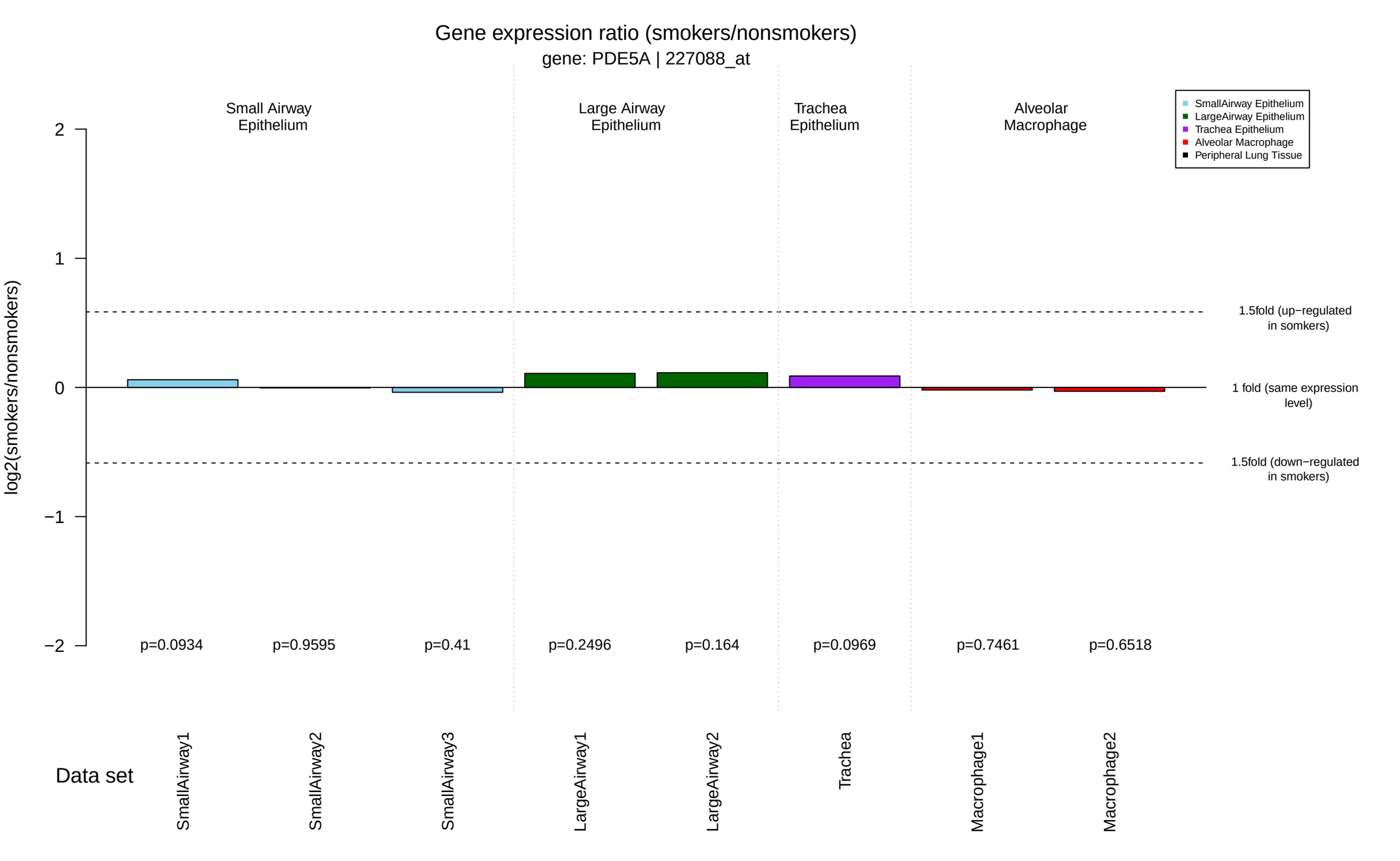
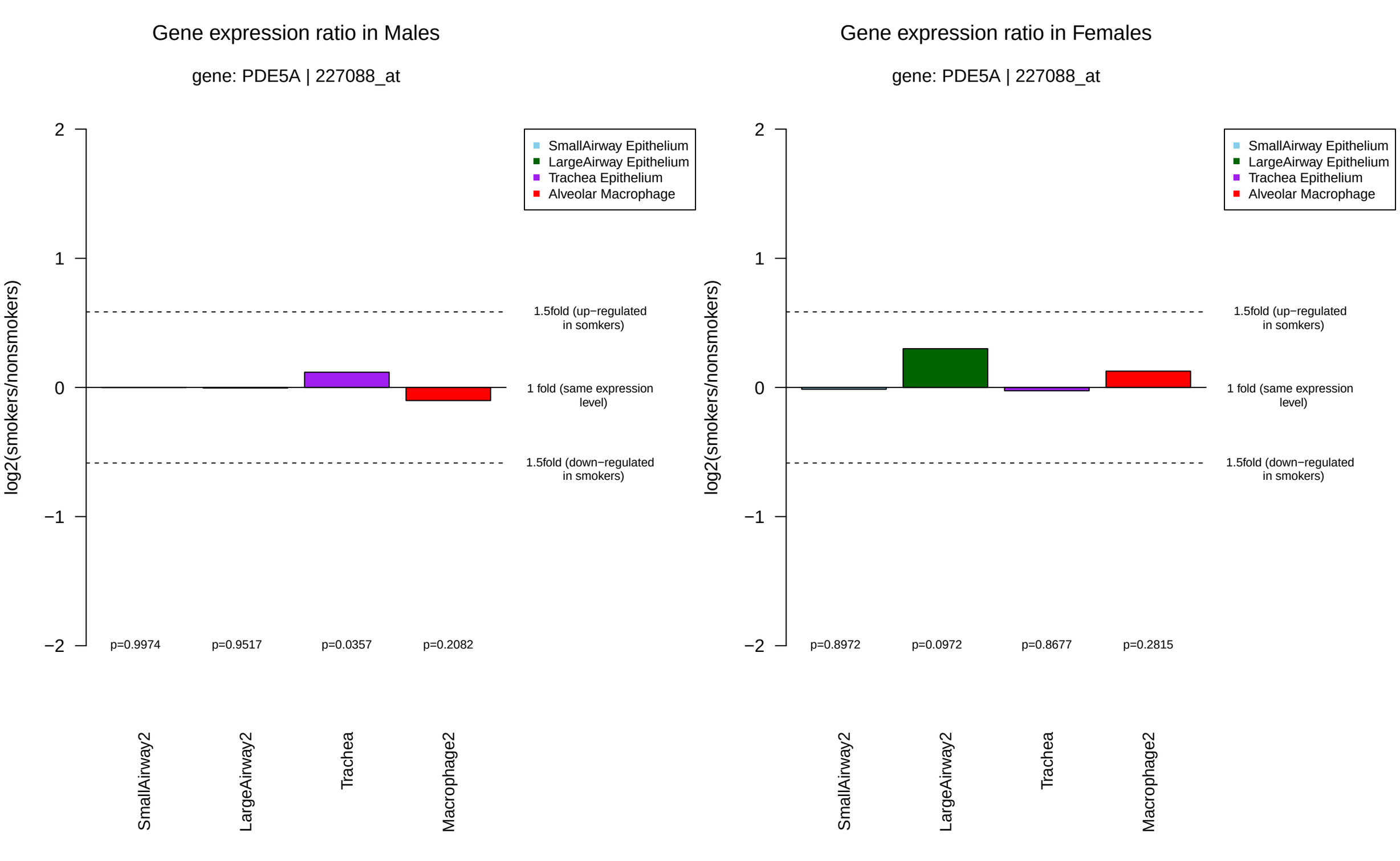
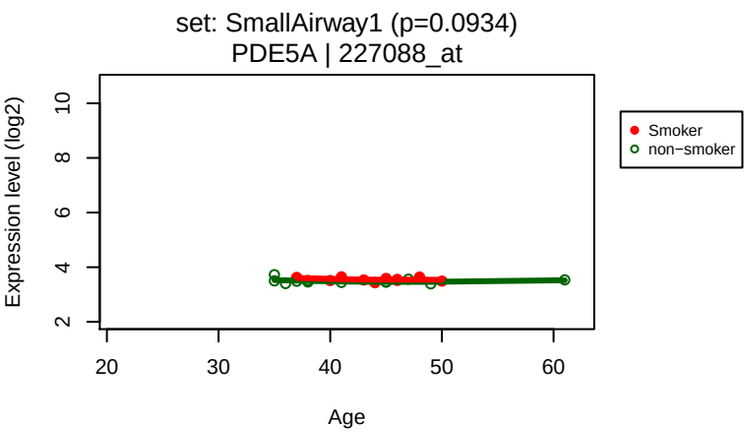
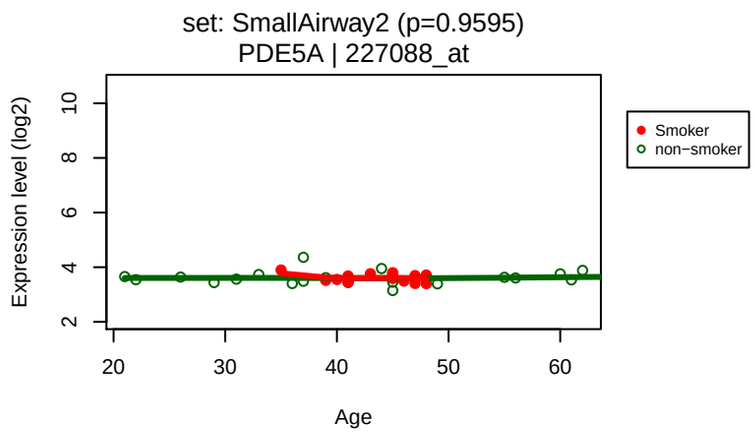
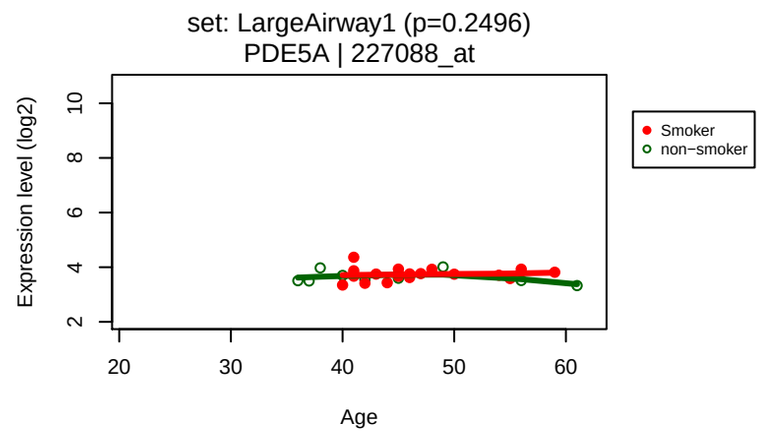

Gene expression of smokers and nonsmokers
gene: PDE5A | 227088_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0934
p=0.9595
p=0.41
p=0.2496
p=0.164
p=0.0969
p=0.7461
p=0.6518
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

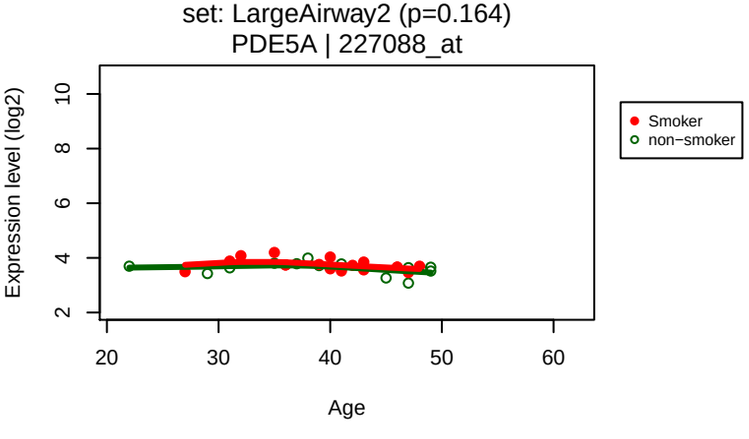

Gene expression ratio (smokers/nonsmokers)
gene: PDE5A | 227088_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0934
p=0.9595
p=0.41
p=0.2496
p=0.164
p=0.0969
p=0.7461
p=0.6518
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

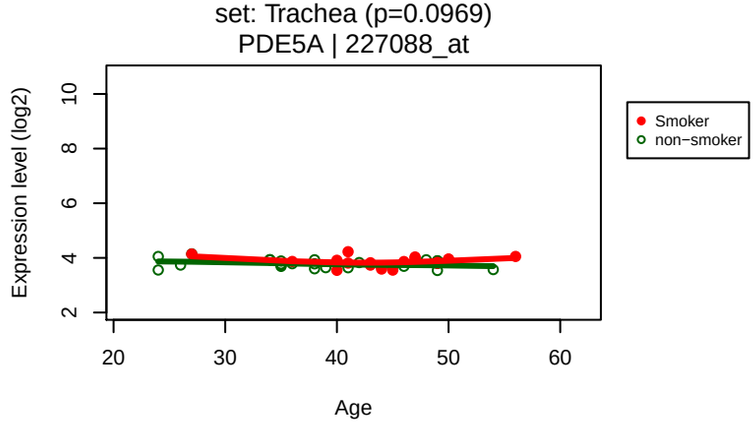

Gene expression ratio in Males
gene: PDE5A | 227088_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9974
p=0.9517
p=0.0357
p=0.2082
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE5A | 227088_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8972
p=0.0972
p=0.8677
p=0.2815
SmallAirway2
LargeAirway2
Trachea
Macrophage2

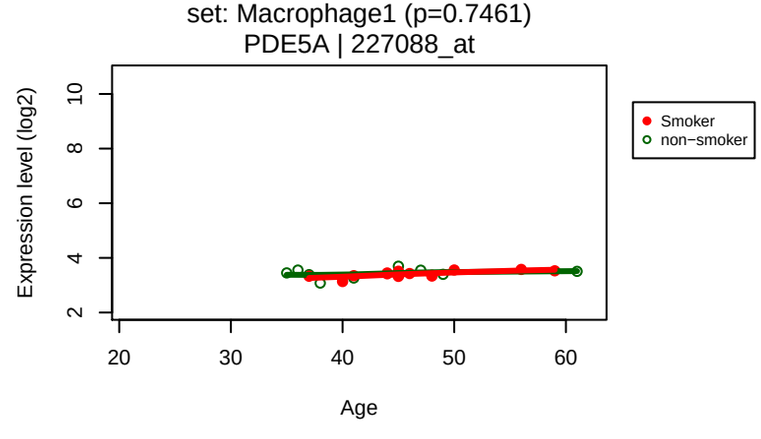

set: SmallAirway1 (p=0.0934)
PDE5A | 227088_at
Expression level (log2)
Age
Smoker
non−smoker

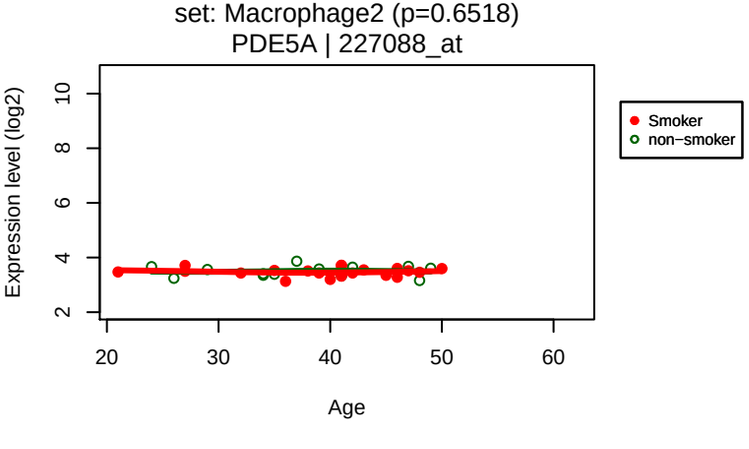

set: SmallAirway2 (p=0.9595)
PDE5A | 227088_at
Expression level (log2)
Age
Smoker
non−smoker

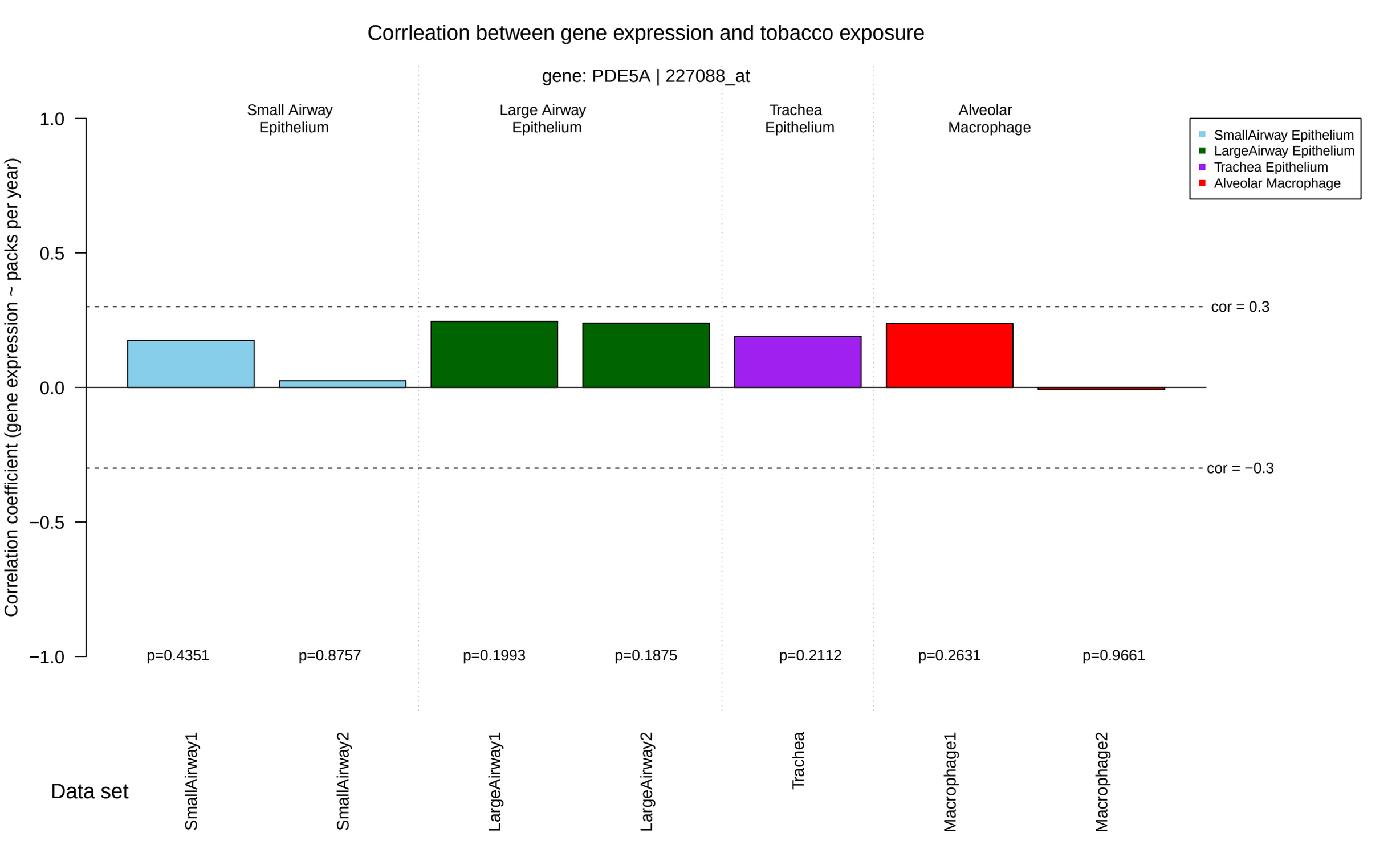

set: LargeAirway1 (p=0.2496)
PDE5A | 227088_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.164)
PDE5A | 227088_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0969)
PDE5A | 227088_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7461)
PDE5A | 227088_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6518)
PDE5A | 227088_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE5A | 227088_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4351
p=0.8757
p=0.1993
p=0.1875
p=0.2112
p=0.2631
p=0.9661
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2